

90 DAYS CHALLENGE

DAY 04 — PYTHON BASICS

Topics: Input • Output • Variable • Local & Global Variable • Expression • Identifier • Keyword

Python Basics: These fundamental concepts help you understand how Python programs take data, store information and execute instructions.

1. INPUT IN PYTHON

Definition	Input means taking data or information from the user while the program is running. Python normally uses the input() function to accept user input.
Syntax	<code>variable = input("Enter something: ")</code>
Example	<code>name = input("Enter your name: ") print(name)</code>
Important Point	The input() function returns the entered data as a string. For numbers, use int() or float() when required.

2. OUTPUT IN PYTHON

Definition	Output is the information displayed by a program on the screen. Python commonly uses the print() function to display output.
Syntax	<code>print(value)</code>
Example	<code>print("Hello Python") print(10 + 5)</code>
Important Point	The print() function can display text, numbers, variables and the results of expressions.

3. VARIABLE

Definition	A variable is a name used to refer to a value stored by a program. Python creates a variable when a value is assigned to a name.
Example	<code>name = "Babli" age = 20</code>
Explanation	Here, name refers to the string "Babli" and age refers to the integer 20.
Important Point	Python is dynamically typed, so you do not need to declare the data type separately before assigning a value.

4. LOCAL VARIABLE

Definition	A local variable is a variable created inside a function and normally available only within that function.
Example	<code>def show(): message = "Hello" print(message)</code>

Explanation	Here, message is local to the function show(). It cannot normally be accessed directly outside that function.
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5. GLOBAL VARIABLE

Definition	A global variable is defined outside functions and can be accessed from different parts of the program, subject to Python's scope rules.
Example	<pre>name = "Python" def show(): print(name)</pre>
Explanation	The variable name is created outside the function, so the function can read it. The global keyword can be used inside a function when you need to rebind a global variable.

6. EXPRESSION

Definition	An expression is a combination of values, variables, operators and/or function calls that Python evaluates to produce a value.
Example	<pre>a + b 10 * 5 x > 10 price * quantity</pre>
Explanation	For example, 10 + 5 is an expression whose result is 15. Expressions can be used wherever Python expects a value.

7. IDENTIFIER

Definition	An identifier is the name used to identify a variable, function, class, module or another program object.
Valid Examples	name, student_age, total2, _count
Rules	1. It can contain letters, digits and underscore (_). 2. It cannot start with a digit. 3. It is case-sensitive: Age and age are different. 4. A Python keyword cannot be used as an ordinary identifier.
Invalid Examples	2name, student-name, class

8. KEYWORD

Definition	A keyword is a reserved word in Python that has a special meaning to the language. A keyword cannot normally be used as the name of a variable, function or class.
Examples	if, else, for, while, def, class, return, True, False
Why Important?	Keywords form the basic grammar and structure of Python programs, such as conditions, loops, functions, classes and exception handling.